Miderm Exam 2 (Take Home)

Connor Mooneyhan

November 21, 2025

2. Let G be a finite group, and let p be a prime integer dividing the order of G. Let P be a Sylow p-subgroup of G, and let H be any subgroup containing $N_G(P)$. Prove that $[G : H] \equiv 1 \bmod p$.

Proof. Write $|G| = mp^n$ where $\gcd(m, p) = 1$. By Sylow's Third Theorem, we know that $[G : N_G(P)] = n_p \equiv 1 \bmod p$. Also, $[G : N_G(P)]$ divides m, and $[G : H]$ divides $[G : N_G(P)]$, so $[G : H]$ divides m. □

5. Find a presentation of a nonabelian group of order $441 = 3^2 \cdot 7^2$ with trivial center whose Sylow 7-subgroup is normal and not cyclic.

Proof. The group $G = (\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}) \rtimes_\varphi \mathbb{Z}/9\mathbb{Z}$ where

$$\phi : \mathbb{Z}/9\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z})$$

satisfies these properties since the Sylow 7-subgroup is $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$, which is not cyclic, and it is normal by construction of the semidirect product, so we just need to show that the center is trivial. □

6. Prove that every group of order $210 = 2 \cdot 3 \cdot 5 \cdot 7$ is not simple.

Proof. We know that n_7 must be a divisor of $2 \cdot 3 \cdot 5$, so it must be 1, 2, 3, 5, 6, 10, 15, or 30. Also, it must be congruent to 1 mod 7, so the only possibilities left are 1 and 15. If $n_7 = 15$, then that accounts for $1 + 15 \cdot 6 = 61$ elements.

Now for n_3, it could either be 1, 2, 5, 7, 10, 14, 35, or 70, but it must be congruent to 1 mod 3, so that leaves us with 1, 7, 10, or 70. Now for n_5, it could either be 1, 2, 3, 6, 7, 14, 21, or 42, but it must be congruent to 1 mod 5, so that leaves us with either 1, 6, or 21.

Every normalizer of a Sylow 7-subgroup necessarily contains that Sylow 7-subgroup, and all the Sylow 7-subgroups intersect trivially, but it's not clear to me what we can say about the intersection of the normalizers. □

7. Prove that a finite group G is nilpotent if and only if every pair of elements with relatively prime order commute.

Proof. We use the equivalent condition for nilpotence that if $|G| = p_1^{n_1} \cdots p_s^{n_s}$ where each p_i is a prime, and $P_i \in \mathrm{Syl}_{p_i}(G)$ for each i, then $G \cong P_1 \times P_2 \times \cdots \times P_s$. If $x \in G$, then for every prime p_i dividing x, there is a non-identity entry in the i^{th} slot representing the group P_i, and likewise the entry is the identity for P_j when p_j does not divide x. If y is relatively prime to x, then they share no primes in common, so given this framing, we see that they will commute. □